ROBERT VOKÁČ, BSc

C++ Systems Engineer - Runtime & Compatibility Layers)
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PROFILE

C++ systems engineer specializing in **runtime architecture, low-level platform layers, and API compatibility frameworks**.

Designed and implemented **custom replacements for XNA, DirectX 3, and WinAPI-style APIs**, enabling real-world games to run without original dependencies across Windows, Linux, Web, and Android.

Experienced in **reverse-engineering behavior, reproducing undocumented semantics, and building cross-platform abstractions over SDL3/OpenGL**.

Transitioning from Java backend development to **full-time C++ systems development**.

Seeking roles in **systems engineering, runtime/framework development, game technology, or platform architecture**.

CORE PROJECTS

CNA - Cross-Platform XNA Runtime

GitHub: github.com/openeggbert/cna

Custom C++ reiimplementatioon of the XNA 4.0 API over SDL3 (~5k LOC), production-tested on real game.

- Designed **XNA-compatible runtime:** game loop, input, rendering, audio, and resource lifecycle
- Implemented **rendering abstraction** supporting SDL_Renderer, OpenGL, and bgfx backends
- Ported **Speedy Blupi (2013)** from C# / XNA to native C++
- Achieved smooth builds on Windows, Linux, Web, and Android from one C++/CNA codebase
- Replaced **MonoGame/.NET/XNA** dependencies with a fully native **C++/SDL3 runtime**
- Reproduced original engine behavior **without access to original engine internals**
- Runtimne performance: **~0.1% CPU usage, ~47 MB RAM**

Playable build: https://speedyblupi.com/SpeedyBlupi2013/

Video: https://www.youtube.com/watch?v=TScL1L-dbHE

Free Direct + Free API - DirectX 3 & WinAPI Compatibility Layer

GitHub: github.com/openeggbert/free-direct · github.com/openeggbert/free-api

Modern C++ reimplementation of DirectX 3 and a WinAPI subset over SDL3 (~6k LOC).

- Implemented **DirectDraw-style 2D rendering and WinAPI event/input handling**
- Built compatibility layer enabling execution of **legacy Windows applications without original DirectX SDK, WinAPI, or runtime dependencies.**
- Supports running:
  - **Speedy Blupi (1998/2001) -** working playable compatibility
  - **Planet Blupi (1996) -** partial compatibility, ongoing debugging

Playable builds:

https://speedyblupi.com/SpeedyEggbert2/

https://speedyblupi.com/PlanetBlupi/

Video: https://www.youtube.com/watch?v=Tl3vNrqgtbo

Hive - Modular Backend Platform

GitHub: github.com/robertvokac/hive

Archived 40k+ LOC modular backend platform.

- Designed **metadata-driven architecture, custom ORM, and REST generation**
- Implemented **plugin system, scheduler, triggers, and validation pipeline**

SKILLS

Languages: C++20/23 (primary), SQL
Systems & Tools: SDL3, OpenGL, CMake, Linux, SQLite, Git

Core Areas:
- Runtime architecture
- API compatibility layers: XNA / DirectX / WinAPI
- Rendering and platform integration
- Abstraction design
- Legacy behavior debugging
- Cross-platform C++ development

EXPERIENCE

Software Engineer – Assist

2017 - present

- Maintained and improved production backend systems using Java and SQL
- Diagnosed and resolved production issues, improving system stability
- Applied long-term system design principles in large legacy codebases
- Developed C++ systems independently, focused on runtime and compatibility layers

Technologies: Java, SQL, Git, Bash

EDUCATION

BSc in Computer Science

Czech University of Life Sciences Prague

2014-2017

LANGUAGES

English: B2
Czech: native